

Week 6, Day 1 — presenter notes

Slides: [week-06-day-01.html](#) → **F** · arrows · **N** notes · **P** print.

Functions and methods inside SMS. Refactor week 5 into `handlers`. No new product feature required.

Timing (3–3.5 hours)

Clock	Block	Slides
0:00–0:15	Health check from week 5	1–2
0:15–0:30	Outcomes + why refactor	3–4
0:30–1:10	Function parts, value, multi-return, defer	5–8
1:10–1:40	Methods, New..., pointer preview, generics light	9–12
1:40–2:20	Live extract handlers + FullName test	13–16
2:20–end	Lab + PR	17–end

Never skip green health after the move. Cut generics live typing first if late.

Live demo

1. Confirm `curl -i :8080/health` is 200.
2. Create `internal/handlers/health.go` + `hello.go` with `New...` and pointer methods.
3. Thin `main` wires `r.GET`.
4. Add `FullName` + `test`; `go test ./internal/handlers`.

Lab gate

- [] Health + hello still 200
- [] Logic in `internal/handlers`
- [] `go test ./internal/handlers` green for `FullName`
- [] PR describes the refactor

Homework

Same gate as PR. Stretch: `internal/routes` register function.

Typical failures

Symptom	Fix
import cycle	handlers must not import main/routes that import handlers wrongly
<code>undefined:</code> <code>handlers</code>	module path + folder under <code>internal/handlers</code>
health JSON changed	copy exact keys
tests not found	<code>go test ./internal/handlers</code> from <code>backend/</code>

What not to teach

Student CRUD, JWT, Postgres, deep pointer theory, interfaces. Tease week 7 only.